Fiscal Year:	FY 2018	**Task Last Updated:**	FY 01/18/2018
PI Name:	Volz, Martin Ph.D.		
Project Title:	Reduction of Defects in Germanium-Silicon (RDGS)		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	256-544-5078
Organization Name:	NASA Marshall Space Flight Center		
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City:	Huntsville	**State:**	AL
Zip Code:	35812	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	96-HEDS-02
Start Date:	03/06/2008	**End Date:**	10/15/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Rogers, Jan	**Contact Phone:**	256.544.1081
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Flight Program:			
Flight Assignment:	ISS NOTE: end date change per P. Downey/MSFC (Ed., 1/12/2015) NOTE: end date change per P. Downey/MSFC (Ed., 12/21/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mazuruk, Konstantin Ph.D. (University of Alabama Huntsville) Cröll, Arne Ph.D. (University of Freiburg)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Original Principal Investigator was Dr. Frank Szofran, Feb 1998-Jan 2004; project continues with Dr. Martin Volz. It is well established that crystals grown without contact with a container have far superior quality to otherwise similar crystals grown in direct contact with a container. In addition to float-zone processing, detached-Bridgman growth is often cited as a promising tool to improve crystal quality, without the limitations of float zoning. In detached growth, most of the melt has contact with the crucible wall, but the crystal is separated from the wall by a small gap, typically 10-100 microns. A liquid meniscus extends between the top edge of the growing crystal and the crucible wall. Detached growth has been found to occur quite often during microgravity experiments and considerable improvements of crystal quality have been reported for those cases. However, no thorough understanding of the process or quantitative assessment of the quality improvements existed prior to the investigation. This project will determine the means to reproducibly grow Ge-Si alloys in the detached mode. Ge-Si alloys are of technological interest because with lower defect density materials it would be possible to make thermal neutron and x-ray optics, faster devices than with silicon, higher efficiency solar cells, and higher thermopower devices. Although these alloys are model materials in the sense that these results should be applicable to many other materials, the major technological significance is that the same production equipment that is currently used for silicon device manufacture can also be used for germanium-silicon devices. This means that using germanium-silicon instead of some other material to achieve improved device performance will save an immense amount of money in development and purchase of production equipment. Several tasks have been completed or are planned as part of the investigation: * measure the relevant material parameters such as contact angle, growth angle, surface tension, and wetting behavior of the GeSi-melt on potential crucible materials; * determine the mechanism of detached growth including the role of convection; * quantitatively determine the differences in defects and impurities among normal Bridgman, detached Bridgman, and float zone (FZ) growth; * investigate the influence of defined azimuthal or meridional flow due to rotating magnetic fields on the characteristics of detached growth; * control time-dependent Marangoni convection in the case of FZ-growth by the use of a rotating magnetic field to examine the influence on the curvature of the solid-liquid interface and the heat and mass transport; and * grow high quality GeSi single crystals with Si concentrations up to 10%. Determine the necessary conditions for reproducibly achieving detachment of Ge-Si alloys in the Bridgman growth method, and quantitatively compare the defect structures and impurity levels of detached Bridgman, normal Bridgman, and floating-zone grown materials.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this research will increase our understanding of the mechanisms which lead to crystal detachment and enhance our ability to control detachment both in microgravity and on Earth. Control of detachment will enable the growth of relatively large diameter crystals with orders of magnitude fewer defects than can be achieved with standard growth techniques. The improvement in material quality may lead to better performance for a variety of electronic devices.
Task Progress:	Experimental work in the reporting period includes the construction of a laboratory crystal growth furnace that can be operated in a rotating magnetic field. The furnace design has been optimized for the traveling heater method or traveling solvent method of crystal growth. In these crystal growth methods, a relatively narrow zone of the starting charge is molten and the molten zone is translated along the charge. The rotating magnetic field interacts with the electrically conducting melt and provides a controllable and noncontact method of stirring the melt. Rotating magnetic fields exist for this purpose for both the Low Gradient Furnace (LGF) and the Solidification and Quench Furnace (SQF) onboard the International Space Station (ISS). Completion of the laboratory furnace is expected before the end of the reporting period. An intriguing possibility in detached growth is to control its occurrence through the adjustment of a parameter external to the growth ampoule. One such possibility is the application of an electric field perpendicular to the growth direction. Electric fields have been shown to have an effect on contact angles and an increase in contact angle will enhance the possibility of detachment. We have developed a microscopic model concerning the electrowetting of an electrically conducting hydrophobic drop on a dielectric substrate. The interaction energy between the drop and substrate includes both van der Waals attractive forces and Born repulsive forces resulting in an equilibrium gap. An augmented Young-Laplace equation is derived and used as the basis for calculations of wetting phenomena both with and without an applied voltage. In the absence of an electric field, a well-defined Young’s angle is established at a distance from the meniscus incipience that is less than 100 times the equilibrium gap. With an electric field applied, the meniscus angle changes continuously from the three-phase line (TPL) until after a distance of at least 10 times the thickness of the dielectric where the Lippmann angle is established. Therefore, the local contact angle is not the Lippmann angle. Rather, as the TPL is approached, the apparent contact angle increases towards the Young’s contact angle value. The results of the model are being prepared for publication.
Bibliography Type:	Description: (Last Updated: 07/09/2021)
Abstracts for Journals and Proceedings	Volz MP, Mazuruk K, Croell A, Sorgenfrei T. "Influence of Containment on the Growth of Germanium-Silicon in Microgravity." 21st American Conference on Crystal Growth and Epitaxy, Santa Fe, NM, July 30-August 4, 2017. 21st American Conference on Crystal Growth and Epitaxy, Santa Fe, NM, July 30-August 4, 2017. Program and abstracts. , Jul-2017

Abstracts for Journals and Proceedings	Mazuruk K, Croell A, Volz MP. "Growth angle –A Microscopic View." 21st American Conference on Crystal Growth and Epitaxy, Santa Fe, NM, July 30-August 4, 2017. 21st American Conference on Crystal Growth and Epitaxy, Santa Fe, NM, July 30-August 4, 2017. Program and abstracts. , Jul-2017